Parameter Efficient Reinforcement Learning from Human Feedback

Mastering Alignment with Frozen Backbones

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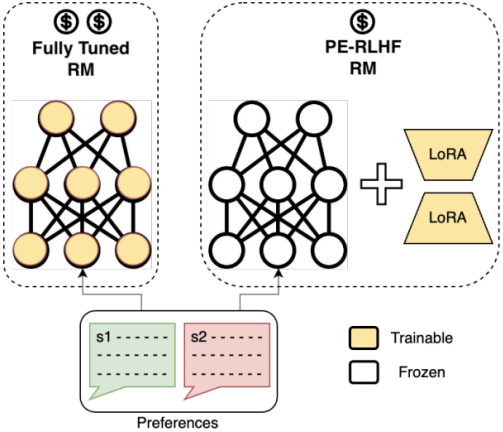
Background: The Building Blocks

Before we solve the problem, let's visualize the two core approaches.

1. **Standard Full Fine-Tuning**
   - Updates **100%** of parameters (Yellow Nodes).
   - High memory cost.
2. **PE-RLHF (LoRA)**
   - Freezes the backbone (White Nodes).
   - Updates only **Adapters** (Trapezoids).
   - Updates <**1%** of parameters.



The Problem: The "Alignment Tax"

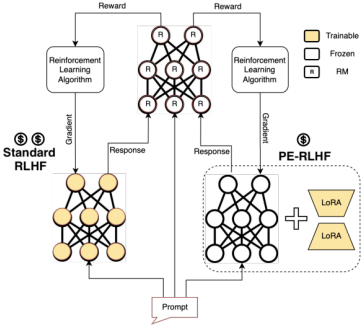
Standard RLHF Bottlenecks:

- **Memory Explosion:** Requires loading **4 distinct full-sized models**:
  1. **Policy** - Generates text
  2. **Value** - Evaluates states
  3. **Reward** - Scores output
  4. **Reference** - Prevents drift
- **Compute Cost:** Gradients stored for billions of parameters.

The Consequence

Fine-tuning 70B+ models is practically impossible on consumer hardware. We need a way to align models without the massive memory overhead.

The Solution: "One Body, Three Hats"



Mechanism: PE-RLHF

- **Frozen Backbone:** The main LLM is *never* updated.
- **LoRA Adapters:** Trainable low-rank matrices injected into **Attention Projection Matrices**.
- **Modular Roles:** Policy, Value, and Reward are just different "adapters" attached to the

The Matrix Trick: Why is it 99% Smaller?

The Math: Instead of training the full update ΔW, we decompose it into two tiny matrices, A and B:

$$W_{new} = W_{frozen} + \underbrace{B \times A}_{\Delta W}$$

The Dimensions (The "Hourglass"):

- W **(Full):** $d \times d$ (e.g., $4096 \times 4096 \approx$ 16M params)
- A **(Compress):** $d \times r$ (e.g., 4096×16)
- B **(Expand):** $r \times d$ (e.g., 16×4096)
- **Total LoRA:** $\approx$ 0.13M params

Why is it still efficient?

The Low-Rank Hypothesis: "Alignment" (learning style/safety) is a much simpler task than "Pre-training" (learning physics/grammar).
We assume the update ΔW has a low "intrinsic rank." We don't need the full capacity of the brain to learn a simple rule like "Don't be toxic."

Result:

- **Smaller:** Store 131k params instead of 16M.
- **Faster:** Compute gradients for only 131k

Understanding Rank (r): The Compression Dial

How do we keep the "Necessary Parts"?

- **The Hypothesis:** Pre-trained models are over-parameterized. The "instruction" to change behavior (e.g., "Be polite") is simple.
- **Intrinsic Dimension:** Imagine a 1000-page book (The Model). To change the genre from Horror to Comedy, you don't rewrite 1000 pages. You just change a few key adjectives.
- **Result:** The "necessary parts" for alignment fit into a tiny subspace (Rank r).

How is r changed and applied?

1. **It is a Hyperparameter:** We choose r *before* training starts (e.g., $r = 4$).
2. **It defines Matrix Shape:**
   - If $d = 4096$:
   - $r = 4$**:** Matrix A is 4096×4. (Tight bottleneck, learns only core rules).
   - $r = 16$**:** Matrix A is 4096×16. (Wider bottleneck, learns nuance).

Key Takeaway: r controls the "Bandwidth" of the adapter. Too small = underfitting; Too large = waste of memory.

Deep Dive: The "Ghost in the Machine"

Critical Question

If the Policy and Value models share the same backbone, how do they think differently?

1. **Physical Storage (Memory)**
   - Only **ONE** copy of the Backbone is stored.
   - **Result:** ~27% Memory Reduction (RL Loop).
2. **Logical Processing (Compute)**
   - **Two Separate Forward Passes.**
   - Pass A: Backbone + Policy Adapter → Action.
   - Pass B: Backbone + Value Adapter → Prediction.

Efficiency Nuance:

- **Reward Model: 90% Faster**.
- **RL Policy: 30% Faster**.
- *Why?* The **Forward Pass Bottleneck**. Generation requires reading the entire frozen backbone.

The Mathematical Objective: The "Leash"

The paper optimizes the standard RLHF objective, but simplifies the calculation of the KL penalty.

$$J(\theta) = \mathbb{E}_{y \sim \pi_\theta} \left[\underbrace{(1-\beta) r_\phi(y|x)}_{\text{Maximize Reward}} - \underbrace{\beta D_{KL}(\pi_\theta^{RL} || \pi^{Anchor})}_{\text{Minimize Drift}} \right]$$

The PE-RLHF Advantage:

- In Standard RLHF, calculating D_{KL} requires running a separate 70B Anchor Model.
- In PE-RLHF, the **Anchor** is simply the **Frozen Backbone** (Adapter Weights $= 0$).
- **Result:** We calculate the drift penalty for free, without loading any extra models.

Experimental Setup: Versatility Stress-Test

Evaluated on **6 datasets** across **3 distinct modalities**:

Task	Dataset	Key Metric
Summarization	Reddit TL;DR *BOLT Messages*	Pairwise Win Rate *Transfer Learning**
Safety **Helpfulness**	Anthropic HH Stanford Human Prefs	Harmless Rate Win Rate
Multimodal Agents	**Visual QA** **UI Automation**	Accuracy vs SFT Action Accuracy

*For **Messages**, they performed Transfer Learning using the Reddit Reward Model.

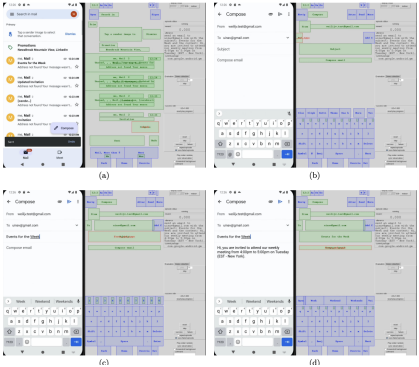
Inside the Judge's Mind: Prompt Engineering

How does the AI Judge know what is "Helpful"? The paper reveals the specific instructions in Appendix Table 8.

Summarization	A good summary is a shorter piece of text that has the essence of the original. It tries to accomplish the same purpose and conveys the key information from the original post. Below we define four evaluation axes for summary quality: coherence, accuracy, coverage, and overall quality. Coherence: This axis answers the question "how coherent is the summary on its own?" A summary is coherent if it's easy to understand when read on its own and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors. Accuracy: This axis answers the question "does the factual information in the summary accurately match the post?" A summary is accurate if it doesn't say things that aren't in the article, it doesn't mix up people, and generally is not misleading. Coverage: This axis answers the question "how well does the summary cover the important information in the post?" A summary has good coverage if it mentions the main information from the post that's important to understand the situation described in the post. A summary has poor coverage if someone reading only the summary would be missing several important pieces of information about the situation in the post. A summary with good coverage should also match the purpose of the original post (e.g. to ask for advice). Overall quality: This axis answers the question "how good is the summary overall at representing the post?" This can encompass all of the above axes of quality, as well as others you feel are important. If it's hard to find ways to make the summary better, the overall quality is good. If there are lots of different ways the summary can be made better, the overall quality is bad. Given a piece of TEXT with SUBREDDIT and TITLE and two of its possible summaries, output 1 or 2 to indicate which summary best adheres to coherence, accuracy, coverage, and overall quality. Text - {text} Summary 1 - {summary-1} Summary 2 - {summary-2} Preferred Summary -

- **Explicit Criteria:** Coherence, Accuracy, Coverage.
- **Role Playing:** "You are an expert rater..."
- This ensures the Reward Signal is aligned with complex human values, not just keywords.

Visualizing the Agent: UI Automation



Task: The agent must sequence actions: *Click Compose* → *Type Email* → *Send*. **Result:** PE-RLHF achieved **86.4% accuracy**, beating the Full Model (81.6%).

Deep Dive: Main Results (Table 1)

		Harmlessness	Helpfulness	Summarization		UI Automation	Visual QA
		Anthropic	SHP	Reddit	Messages	UI Automation	VQAv2
RM *	Full-tuning	76.56%	**83.2%**	78.7%	-	**93.12%**	-
	PE-RLHF	**78.71%**	82.2%	**79.7%**	-	91.8%	+0.5%†
Efficiency	PE HBM	43.1%	48%	50%	-	56%	74%
	Speed-up	1.6×	1.5×	1.7×	-	1.9×	1.15×
RL **	Full-tuning	96.6%	**63.0%**	**87%**	73.2%	81.6%	**+5.5%**‡
	PE-RLHF	**98.2%**	61.3%	86.5%	**75.5%**	**86.4%**	+3.9%‡
Efficiency	PE HBM	80%	75%	75%	80%	74%	74%
	Speed-up	1.3×	1.15×	1.15×	1.05×	1.20×	1.24×

* Accuracy.
** Win rate by a Judge model.
† Absolute win rate change compared to Supervised Fine-Tuning baseline.
‡ Absolute accuracy change compared to Fully-Tuned baseline.

Unpacking the Results: What do the numbers mean?

1. The "Judge" (Reward Model)

Performance: Better or Equal

- **Harmlessness:** PE-RLHF (78.7%) actually *outperformed* Full Tuning (76.6%).
- **Implication:** A small adapter is sufficient to learn human preferences.

Efficiency: Massive Gains

- **Speed:** Training is nearly **2x faster** (1.9x).
- **Memory:** Cuts footprint in **half**.

2. The "Agent" (RL Policy)

Performance: Surprising Wins

- **Safety:** Reached **98.2%** harmlessness (vs 96.6%).
- **Agents:** UI Automation accuracy jumped to **86.4%** (vs 81.6%).

Efficiency: Moderate Gains

- **Speed:** Only **1.3x faster**.
- **Constraint:** The generation step (forward pass) is the bottleneck, preventing larger speedups.

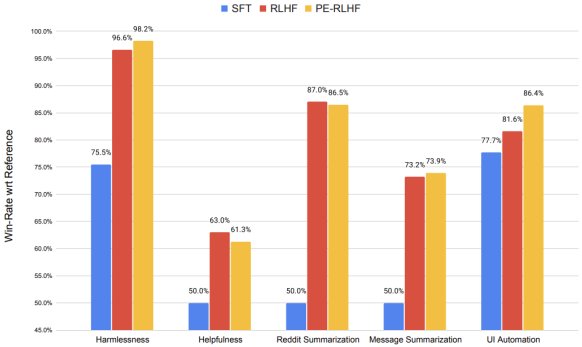
Defending the Metric: Can we trust the AI Judge?

A common critique is reliance on "AI Judges." The authors validated this against human annotators (Appendix A.4).

Task	Agreement rate
Summarization	77.2%
Harmless Response Generation	90.7%

- **Harmlessness: 90.7% Agreement** between Human and AI Judge.
- **Summarization: 77.2% Agreement**.
- **Implication:** The AI Judge is a statistically valid proxy for human preference.

Visualizing Success: Win Rates



Takeaway: The Yellow Bars (PE-RLHF) consistently match or exceed the Red Bars (Full RLHF).

Critical Insight: The VQA Limitation

Visual QA Performance (vs. SFT Baseline)

Method	Improvement over SFT
Full Fine-Tuning	**+5.5%**
PE-RLHF	+3.9%

Analysis:

- This is the **only** task where PE-RLHF notably lagged behind Full Tuning.
- **Reasoning:** Complex Vision-Language reasoning requires updating deep weights.
- **Conclusion:** Freezing the backbone creates a "capacity ceiling" for multimodal tasks.

Ablation Studies: What Matters?

Model	Setting	Harmlessness	Helpfulness	Summarization	UI Automation
		Anthropic	SHP	Reddit	UI Automation
PaLM 2 S	Fully-Tuned	76.6%	83.2%	78.7%	93.1%
	LoRA 16	75.0%	81.3%	77.0%	92.2%
	LoRA 8	79.1%	81.6%	77.3%	92.0%
	LoRA 4	78.7%	82.2%	79.7%	91.2%
	LoRA 1	76.0%	80.9%	77.2%	90.8%
PaLM 2 XS	Fully-Tuned	77.0%	82.0%	78.1%	88.6%
	LoRA 8	76.8%	82.6%	-	89.2%
	LoRA 4	-	81.8%	76.8%	89.7%
PaLM 2 XXS	Fully-Tuned	73.4%	80.2%	75.4%	84.4%
	LoRA 8	72.5%	79.3%	-	87.1%
	LoRA 4	-	77.8%	73.2%	85.8%

Model	Setting	Visual Question Answering
		VQAv2*
Gemini Pro	Fully-Tuned	0.0%
	LoRA 32	-1.2%
	LoRA 16	-1.0%
	LoRA 8	-1.1%
	LoRA 4	-0.6%
	LoRA 1	-1.0%

* Absolute accuracy change compared to Fully-Tuned baseline.

1. **"Bigger is Better"**
   - PE-RLHF matches Full Tuning on **Large** models (PaLM 2 S).
   - It struggles slightly on **Tiny** models (XXS).
2. **Rank Sensitivity**
   - **Reward Models:** Work fine with **Rank 4**.
   - **Policy Models:** Benefit from **Rank 16+**.

Engineering Rigor: Hyperparameter Sweeps

The performance wasn't accidental. The authors conducted extensive sweeps (Appendix A.3).

LLM Setting	DR	LR
S	0.01	1e-5
S LoRA Rank 1	0.01	2e-4
S LoRA Rank 4	0.01	1e-4
S LoRA Rank 8	0	2e-4
S LoRA Rank 16	0.01	5e-5
XS	0.01	5e-5
XS LoRA Rank 4	0.05	1e-4
XXS	0	1e-5
XXS LoRA Rank 16	0.05	1e-4

Critical Finding: Learning Rates

- **Full Fine-Tuning:** 1×10^{-5} (Conservative).
- **LoRA Tuning:** 2×10^{-4} (**20x Higher**).

Why is LoRA aggressive?

1. **Zero Initialization:** LoRA matrix B starts at 0. The adapter output is initially null.
2. **The "Kick":** Unlike pre-trained weights which just need nudging, LoRA adapters need a large gradient update to break away from zero and start affecting the

Conclusion

- **Conceptually:** We can dismantle the RLHF bottleneck by treating Policy, Value, and Reward as modular LoRA adapters.
- **Empirically:**
  - **Parity** achieved on 5 out of 6 tasks.
  - **Memory** reduced by 50% (RM) and 27% (RL).
  - **Speed** increased by 90% (RM).
- **Impact:** Domain-specific alignment is now accessible without industrial-scale compute clusters.

Future Directions: Where do we go from here?

The authors identify two major limitations to address in future work (Section 7):

1. **The Generalization Problem:**
   - PE-RLHF works well in-domain but may overfit.
   - **Proposed Solution: Mixture-of-LoRA (MoLE)**. Use different adapters for different tasks and route dynamically.
2. **The Reward Hacking Problem:**
   - Efficient models are still prone to "gaming" the Reward Model.
   - **Proposed Solution: Weight Averaging (WARM)**. Average the weights of several best checkpoints to create a more robust policy.

Ethical Considerations: The Double-Edged Sword

Democratization Risk

"While reducing alignment barriers holds immense promise... it also presents ethical challenges."

The Benefit:

- Allows academic labs to study safety alignment.
- Enables personalized agents on consumer hardware.

The Risk:

- **Malicious Alignment:** Bad actors can now "align" models to be harmful using cheap GPUs.
- **Mitigation:** Requires "White-box" access controls and robust monitoring.